

Project 1: Language Translation Transformer

Content	
Project Title	Language Translation Transformer
Category	GenAI
Problem Statement	This project leverages a Transformer model which is built from scratch to translate text from English to Telugu. It uses a large parallel corpus of English-Telugu sentences to train the model, ensuring coverage of diverse linguistic nuances. The attention mechanism in Transformers aids in capturing context and handling long-range dependencies, which is crucial for translating complex sentence structures. Additional tokenization and transliteration layers help correctly map Telugu script characters and preserve meaning across language boundaries. Ultimately, the goal is to provide fast, accurate translations that enhance digital communication and knowledge sharing for Telugu speakers
Tech Stack	Transformers , MERN
Max Groups	3

Project 2: RAG for Legal Question Answering and Summarization.

Content	
Project Title	RAG for Legal Question Answering and Summarization.
Category	GenAI
Problem Statement	This project is a Retriever-Augmented Generation (RAG) system tailored for legal documents. It retrieves the most relevant legal text from an indexed repository and then uses a large language model (Legal BERT)to generate context-rich, accurate responses.
Tech Stack	Transformers , MERN
Max Groups	3

Project 3: SafeStreet - Road Damage Detection System

Content	
Project Title	SafeStreet - Road Damage Detection System
Category	GenAI
Problem Statement	This project leverages a Vision Transformer (ViT) to automate damage identification and prioritization from uploaded images. First, the ViT classifies the type of damage (e.g., cracks, dents, corrosion) and assesses its severity, drawing on learned features that capture subtle visual details. Next, the system generates a concise text summary that documents the specific damage type, severity level, and recommended repair priority. This integrated approach streamlines inspection workflows, cuts down manual evaluation, and delivers consistent, data-driven insights particularly useful for industries like insurance, automotive, construction, or manufacturing. By combining powerful vision models with intelligent text generation, the project ensures both accurate visual analysis and clear, actionable reporting for damage assessment.
Tech Stack	MERN, Vision Transformers
Max Groups	3

Project 4: SceneSolver - A Smart Tool for Crime Scene Investigation

Content	
Project Title	SceneSolver - A Smart Tool for Crime Scene Investigation
Category	GenAI
Problem Statement	SceneSolver is an AI-powered forensic platform that automates crime scene analysis using CLIP and Vision Transformers (ViT). The platform processes images to identify crime types, extract key visual evidence, and generate detailed crime scene summaries. By supporting batch processing, SceneSolver enhances the efficiency of forensic investigations, enabling professionals to analyze large amounts of visual data quickly and accurately.
Tech Stack	CLIP , Vision Transformers , MERN
Max Groups	3

Project 5: Stale Fruit Detection Using Swin Transformer

Content	
Project Title	Stale Fruit Detection Using Swin Transformer
Category	GenAI
Problem Statement	The project focuses on using the Swin Transformer, a state-of-the-art deep learning model, for the automated detection of stale fruits in cold storage. The goal is to create a system that can accurately classify fruits as fresh or stale based on their visual appearance, leveraging the power of transformers to capture both local and global features. This system will improve inventory management, reduce food waste, and streamline the monitoring process in cold storage facilities.
Tech Stack	PyTorch , Vision Transformer, CNNs, PIL , Flask , MERN, FastAPIs
Max Groups	3

Project 6: Transpolymer

Content	
Project Title	Transpolymer
Category	GenAI
Problem Statement	TransPolymer: a Transformer-based language model for polymer property predictions
Tech Stack	Transformers , MERN
Max Groups	3